

AI Browser Extension for Phishing Detection

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1. Introduction

This project proposes the development of an AI-powered browser extension that detects phishing websites in real time while users browse the internet. The extension analyzes the structure and characteristics of the URL and webpage to determine whether the site is likely to be malicious. If a phishing risk is detected, the system warns the user immediately through a browser alert.

2. How the System Works

The system consists of three major components: URL analysis, machine learning classification, and a browser extension interface.

First, when a user visits a website, the browser extension extracts relevant features from the URL.

Second, the extracted features are sent to a machine learning model trained on a phishing website dataset. The model analyzes the features and predicts whether the website is legitimate or potentially malicious.

Finally, the extension receives the prediction result and notifies the user if the site is considered suspicious. The warning message informs the user about the possible phishing attempt and advises them to avoid entering sensitive information.

3. Objectives

The main objectives of this project are:

- To design and develop a browser extension capable of detecting phishing websites in real time.
- To build and train a machine learning model that can accurately classify phishing URLs.
- To protect users from phishing attacks by providing immediate warnings.
- To demonstrate the practical application of artificial intelligence in cybersecurity.
- To create an interactive and user-friendly security tool that enhances safe browsing.